THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 7 — SCIENCE · MEDIUM LEVEL UNIT TEST SET 1

Chapters 1–4: **Exploring Substances: Acidic, Basic and Neutral, Electricity: Circuits and their Components, The World of Metals and Non-metals, Changes Around Us: Physical and Chemical** | RBSE Pattern | Session 2026–27

Time: 2 Hours Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–15 are Objective Type (1 mark each).
3. Q. 16–20 are Very Short Answer Type (2 marks each).
4. Q. 21–25 are Short Answer Type (3 marks each).
5. Q. 26–27 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 24–25 and Q. 26–27.
7. Write the question number before each answer.
8. Medium level: multi-step questions.

SECTION — A (Objective Type Questions) 15 × 1 = 15

Choose the correct option and write it in your answer book.

1. A solution turns both red and blue litmus paper to red. The solution is: 1

(a) neutral **(b)** basic **(c)** salty **(d)** acidic

2. Four cells of 1.5 V each are joined in series. The total voltage is: 1

(a) 4.5 V **(b)** 3.0 V **(c)** 6.0 V **(d)** 1.5 V

3. A substance is shiny, can be hammered into thin sheets and conducts electricity. It is probably a: 1

(a) gas **(b)** liquid non-metal **(c)** non-metal **(d)** metal

4. Which one of these pairs contains only chemical changes? 1

(a) burning of coal, ripening of fruit **(b)** cutting paper, tearing cloth

(c) dissolving sugar, freezing water **(d)** melting wax, boiling water

5. In $\mathrm{HCl + NaOH \longrightarrow NaCl + H_2O}$, the base is: 1

(a) HCl **(b)** $\mathrm{H_2O}$ **(c)** NaOH **(d)** NaCl

6. Which of the following is not a safety measure? 1

(a) using damaged wires **(b)** using MCB

(c) using earth wire **(d)** using insulated tools

7. When magnesium ribbon is burnt and the ash dissolved in water, the solution turns: 1

(a) blue litmus red **(b)** green **(c)** colourless **(d)** red litmus blue

8. In the experiment with three nails, the nail in dry air does not rust because: 1

(a) no moisture **(b)** no sunlight **(c)** no iron **(d)** no oxygen

9. Why does vinegar taste sour? 1

(a) it contains acetic acid **(b)** it contains sugar

(c) it contains sodium **(d)** it contains salt

10. A bulb uses a battery of two cells. If a third cell is added in series, the bulb: 1

(a) becomes dimmer **(b)** becomes cold **(c)** goes off **(d)** becomes brighter

11. Why does an aluminium vessel not corrode easily? 1

(a) it is brittle **(b)** it is sonorous

(c) a protective oxide layer forms **(d)** it is a non-metal

12. Which statement about crystallisation is true? 1

(a) crystals form from a gas always
(b) crystals form by burning
(c) crystals form when a hot saturated solution cools
(d) crystals form by sieving

13. A solution that turns red litmus blue and turmeric red is: 1

(a) basic **(b)** neutral **(c)** sugary **(d)** acidic

14. Two cells of 1.5 V are connected with opposite polarity in series. The net voltage is about: 1

(a) 0 V **(b)** 3 V **(c)** 4.5 V **(d)** 1.5 V

15. A substance is dull, brittle and does not conduct electricity. It is probably a: 1

(a) metal **(b)** non-metal **(c)** alloy **(d)** magnet

SECTION — B (Very Short Answer Type) — 5 × 2 = 10

Answer in about 20–30 words each.

16. A solution turns red litmus blue. What does it contain? Give an example. 2

17. A student builds a tester with a cell, bulb and two wires. The bulb glows when a coin is placed but not with a plastic cap. Explain. 2

18. Why do silver articles turn black on keeping in air? 2

19. Why is the burning of a candle called both a physical and a chemical change? 2

20. Write a neutralisation reaction of sulphuric acid with sodium hydroxide. 2

SECTION — C (Short Answer Type) — 5 × 3 = 15

Answer in about 40–60 words each.

21. A solution X turns red cabbage extract green and turmeric red. Another solution Y turns it red and turmeric stays yellow. Identify X and Y. 3

22. Four 1.5 V cells are joined in series to light a 6 V bulb. Does the bulb glow normally? Explain. 3

23. Predict what will happen if a copper nail is dipped in iron sulphate solution and explain. 3

24. You are given a piece of iron nail and a copper sulphate solution. Predict the changes and say whether the change is physical or chemical. 3

OR

Why are some stomach pills chewed and others dissolved? 3

25. Describe how an MCB works in simple terms. 3

OR

Explain why a tarnished copper vessel can be cleaned with lemon or tamarind. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

26. You are given five unlabelled liquids: vinegar, lime water, sugar solution, lemon juice and soap solution. Design a test to identify each. 5

OR

Describe a complete experiment to find which materials are conductors, including the circuit diagram in words, table of results and conclusion. 5

27. Plan an investigation to classify ten given materials as metals or non-metals using lustre, hardness, conduction and sound. 5

OR

Plan an investigation to find the effect of salt water and plain water on rusting of iron nails, with the set-up, observations and conclusion. 5

END OF QUESTION PAPER **Total: 50 Marks**

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